

20250908EB_MN (RNA, virus titer test)

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- 2025年9月8日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年9月10日 星期三
- T2 -> T3
- 2025年9月12日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年9月15日 星期一
- T4 -> T5
- 2025年9月17日 星期三
- T5 -> T6
- 2025年9月19日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年9月22日 星期一
- T6 -> T7
Dissociation
 - Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) over weekend
 - Remove the PLO and wash the plate with PBS once
 - Coat the plate with laminine for 1hr.
 - Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 - Add 2mL accutase and tap the cells
 - Put the tube into 37C water bath. Flick the tube every 5min until 50min, but the EBs looked very intact. I pipetted it up and down for 30 times, but the EBs were still there. I transfer the supernatant to process the following steps and add 1mL new accutase to remaining EBs.
 - Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.
 - Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 200rcf, 3.5min. Remove the supernatent. Add 1mL of T7 and pipette it gently and thoroughly.
11. Add 4mL more T7 to resuspend the cells and count it.
12. The yield was too low, so I processed batch 2 after 30min incubation of new 1mL accutase -> 1ml HI-FBS and CTWM -> filter -> centrifuge
13. Plate the cells with T7.

cell info ^

	A	B
1	cell	passage
2	WT11	

Table1 ^

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

Plate A

^

	1	2	3
A			
B			

Plate B

^

	1	2	3	4	5	6
A				vLH 78 2uL		
B				vLH78 10uL		
C				vLH103 1uL		
D				vLH103 5uL		

- 2025年9月23日 星期二
- Change T7 -> T7 for plate B
- The coating was old and have been dried out, so the plating was bad. Especially in the middle of A1 and D1.
- 2025年9月24日 星期三
- Plate A: T7 -> T7
- 2025年9月25日 星期四
- T7 -> BP T8
- 2025年9月27日 星期六
- 1/2 T8 -> 1/2 T8
- 2025年9月29日 星期一
- 1/2 T8 -> 1/2 T8
- 2025年9月30日 星期二
- Collect 100uL media from Plate A A1 and A2.
- 2025年10月1日 星期三
- 1/2 T8 -> 1/2 T8
- 2025年10月2日 星期四
- Plate A
- Treat cells for 24h at 5:20-23pm (rinse the cells once)

Well1 ^

	1	2	3
A	2.5mM gluc 1h	2.5mM gluc 24h	
B	0.15mM gluc 1h	0.15mm gluc 24h	

2025年10月3日 星期五

Plate A

Observation: The cells have more non-MNs on the plate for this batch. Many clusters in the middle have non-MNs beneath.

Treat cells for 1h at 4:21-25pm (rinse cells once)

-
1. Collect RNA at 5:20pm with Trizol

2. Wash cells with 1mL cold PBS

3. Remove PBS

4. Add 1mL Trizol for each well. Incubate in RT for 5min

5. Collect the Trizol into 1.5mL eppendorf

6. Freeze samples in -80.

Plate B

Check the virus expression on PlateB on old spinning disk confocal

60x, 704x704 ^

	A	B	C	D
1		vLH78		vLH103
2	2uL	There are decent number of cells expressing good nucGEMs, but some cells have too dense GEMs	1uL	Good # of GEM+ cells. GEM desity is good. (some are a bit more, but okay, especially under 1.5x zoom)
3	10uL	More GEM+ cells, but also more cells with higher GEM density. Still can find some good cells	5uL	good # of GEM+ cells, Most cells have food GEM density, but there are more cells with too many GEMs.